

MGS 2026 — Category Theory

Exercises

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1 Sets and Warm-Up

The exercises in this section use only sets and functions; no category theory yet. Notation: $f: X \rightarrow Y$ for a function; $\text{id}_X: X \rightarrow X$ for the identity; $g \circ f$ for composition (apply f first).

Throughout, $\mathbb{N}$ denotes the natural numbers $\{0, 1, 2, \dots\}$ with successor $s(n) = n + 1$.

Exercise 1.1 (Isomorphisms) Let $f: X \rightarrow Y$ be a function.

- (a) Show that if $g, h: Y \rightarrow X$ both satisfy $g \circ f = \text{id}_X$, $f \circ g = \text{id}_Y$ and $h \circ f = \text{id}_X$, $f \circ h = \text{id}_Y$, then $g = h$. Conclude: the two-sided inverse f^{-1} is unique when it exists.
- (b) Show that f has a two-sided inverse if and only if f is a bijection.
- (c) Prove: if $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are isomorphisms, so is $g \circ f$, and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
- (d) Prove: id_X is an isomorphism and $\text{id}_X^{-1} = \text{id}_X$.

Exercise 1.2 (Terminal sets) A set T is called *terminal* if for every set X there is exactly one function $X \rightarrow T$.

- (a) Show that T is terminal if and only if T has exactly one element (i.e. T is a singleton in the usual sense).
- (b) Show that any two terminal sets are isomorphic, without appealing to (a).

Exercise 1.3 (Initial sets) A set I is called *initial* if there is exactly one function $I \rightarrow X$ for every set X . Verify that a set is initial if and only if it is empty.

Exercise 1.4 (Kuratowski pairing and Cartesian product) The *Kuratowski definition* sets $(x, y) := \{\{x\}, \{x, y\}\}$ and $X \times Y := \{(x, y) \mid x \in X, y \in Y\}$ with $\text{fst}(x, y) = x$, $\text{snd}(x, y) = y$.

- (a) Show that $(x, y) = (x', y')$ implies $x = x'$ and $y = y'$. (*Hint*: consider the cases $x = y$ and $x \neq y$ separately.)
- (b) Prove the universal property: for every set Z and functions $f: Z \rightarrow X$, $g: Z \rightarrow Y$, there exists a unique $h: Z \rightarrow X \times Y$ with $\text{fst} \circ h = f$ and $\text{snd} \circ h = g$.

Exercise 1.5 (Monomorphisms and epimorphisms) A function $f: X \rightarrow Y$ is called a *monomorphism* (mono) if for all sets Z and functions $g, h: Z \rightarrow X$, $f \circ g = f \circ h$ implies $g = h$; and an *epimorphism* (epi) if for all sets Z and functions $g, h: Y \rightarrow Z$, $g \circ f = h \circ f$ implies $g = h$.

- (a) Show f is a mono iff f is injective.
- (b) Show f is an epi iff f is surjective.

Now let $(P, \leq)$ and $(Q, \leq)$ be posets and $f: P \rightarrow Q$ a monotone map. Call f a *mono in Pos* (resp. *epi in Pos*) if the same conditions hold with all sets and functions replaced by posets and monotone maps.

- (c) Show f is a mono in **Pos** iff f is injective.
- (d) Show f is an epi in **Pos** iff f is surjective. (*Hint*: for the non-trivial direction, suppose $q_0 \notin f(P)$ and take the discrete poset $\{0, 1\}$ as codomain.)

An epi $f: P \rightarrow Q$ in **Pos** *splits* if there exists a monotone map $g: Q \rightarrow P$ with $f \circ g = id_Q$.

- (e) ($\star$) Let $P = \{a, b, c\}$ with $a < b$ and c incomparable to both, and let $Q = \{0 < 1 < 2\}$. Define $f: P \rightarrow Q$ by $f(a) = 0$, $f(b) = 1$, $f(c) = 2$. Show f is surjective (hence epi by (d)), but does not split.

Exercise 1.6 (Structural recursion on $\mathbb{N}$) For any set A , element $a \in A$, and function $h: A \rightarrow A$, there is a unique $f: \mathbb{N} \rightarrow A$ with

$$f(0) = a, \quad f(n+1) = h(f(n)).$$

- (a) Use induction to show that such f is unique.
- (b) Define **double**: $\mathbb{N} \rightarrow \mathbb{N}$ by structural recursion and prove **double**(n) = $2n$ by induction.
- (c) Show that *strong recursion* is derivable from structural recursion: for any set X , function $g: X \rightarrow A$, and function $h: A \times X \rightarrow A$, there is a unique $f: \mathbb{N} \times X \rightarrow A$ with

$$f(0, x) = g(x), \quad f(n+1, x) = h(f(n, x), x).$$

(*Hint*: run recursion over A^X as A .)

- (d) Using strong recursion, define addition of two numbers **add**: $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$.
- (e) Show that primitive recursion is derivable from strong recursion: for any set X , function $g: X \rightarrow A$, and function $h: A \times \mathbb{N} \times X \rightarrow A$, there is a unique $f: \mathbb{N} \times X \rightarrow A$ with

$$f(0, x) = g(x), \quad f(n+1, x) = h(f(n, x), n, x).$$

(*Hint*: run strong recursion over $A \times \mathbb{N}$, instead of A .)

- (f) Using primitive recursion, define **factorial**: $\mathbb{N} \rightarrow \mathbb{N}$.

Recall that $\text{List}(X)$ denotes the set of finite lists over a set X , with empty list $[]$ and cons $x :: xs$.

Exercise 1.7 (Structural recursion on $\text{List}(X)$) For any set A , element $a \in A$, and function $h: X \times A \rightarrow A$, there is a unique $f: \text{List}(X) \rightarrow A$ with

$$f([]) = a, \quad f(x :: xs) = h(x, f(xs)).$$

- (a) Define **length**: $\text{List}(X) \rightarrow \mathbb{N}$ by choosing appropriate a and h .
- (b) Define concatenation $++$: $\text{List}(X) \times \text{List}(X) \rightarrow \text{List}(X)$ by recursion on the first argument.
- (c) Prove by induction: **length**($xs ++ ys$) = **length**(xs) + **length**(ys).
- (d) ($\star$) State and prove the analogue of primitive recursion for $\text{List}(X)$: a recursion scheme where the step function also has access to the tail xs , not just the head x and the recursive result $f(xs)$.

2 Categories and Functors

Throughout, verify all stated facts carefully: check identity laws and associativity for categories; check preservation of composition and identities for functors.

Exercise 2.1 (Verifying categories) Show that each of the following is a category.

- (a) **Set**: sets and functions, composition of functions.
- (b) **Pos**: posets and monotone maps.
- (c) **Mon**: monoids and monoid homomorphisms.
- (d) **Rel**: sets and relations, where the composite of $R \subseteq X \times Y$ and $S \subseteq Y \times Z$ is $S \circ R = \{(x, z) \mid \exists y. (x, y) \in R, (y, z) \in S\}$.

Exercise 2.2 (Monoid as one-object category) Let $(M, \cdot, e)$ be a monoid. Define a category $\mathbf{BM}$ with a single object $\bullet$ and $\text{Hom}(\bullet, \bullet) = M$, with composition $g \circ f = g \cdot f$ and identity $\text{id}_\bullet = e$.

- (a) Verify the category axioms.
- (b) Show that functors $\mathbf{BM} \rightarrow \mathbf{BN}$ correspond exactly to monoid homomorphisms $M \rightarrow N$.
- (c) What are the isomorphisms in $\mathbf{BM}$? Relate to invertible elements of M .

Exercise 2.3 (Poset as category) Let $(P, \leq)$ be a poset, viewed as a category (objects = elements, unique morphism $x \rightarrow y$ iff $x \leq y$).

- (a) What are the isomorphisms in this category?
- (b) What are the terminal and initial objects (if they exist)?
- (c) What is the opposite category $(P, \leq)^{\text{op}}$?
- (d) A functor $(P, \leq) \rightarrow (Q, \leq)$ is exactly a ... ? Prove it.

Exercise 2.4 (Isomorphisms in a category) Let $\mathcal{C}$ be a category.

- (a) Show that any morphism has *at most one* two-sided inverse.
- (b) Show that the relation $A \cong B$ (there exists an isomorphism $A \rightarrow B$) is an equivalence relation on objects.
- (c) Suppose $f: A \rightarrow B$ has a left inverse g (i.e. $g \circ f = \text{id}_A$) and a right inverse h (i.e. $f \circ h = \text{id}_B$). Show $g = h$ and hence f is an isomorphism.

Exercise 2.5 (Products)

- (a) Verify the universal property of Cartesian product in **Set**: for any set Z and functions $f: Z \rightarrow X, g: Z \rightarrow Y$ there is a unique $h: Z \rightarrow X \times Y$ with $\pi_X \circ h = f$ and $\pi_Y \circ h = g$.
- (b) Show that any two products of X and Y in a category are isomorphic via a unique isomorphism that commutes with the projections.
- (c) Verify the axioms of binary products: $\pi_1 \circ \langle f, g \rangle = f, \pi_2 \circ \langle f, g \rangle = g, \langle \pi_1 \circ h, \pi_2 \circ h \rangle = h$.
- (d) Show that a category has all finite products iff it has a terminal object and binary products.

Exercise 2.6 (Coproducts)

- (a) State the definition of coproduct by dualising the definition of product (reverse all arrows).
- (b) Verify that the disjoint union is a coproduct in **Set**.
- (c) In **Pos**: describe the coproduct of two posets $(P, \leq)$ and $(Q, \leq)$.
- (d) $(\star)$ Show that the coproduct of monoids $(M, \cdot, e_M)$ and $(N, \cdot, e_N)$ in **Mon** is *not* the disjoint union. (Hint: the coproduct is the free product $M * N$, the monoid of alternating words over M and N .)

Exercise 2.7 (Opposite category) Let $\mathcal{C}$ be a category.

- (a) Verify that $\mathcal{C}^{\text{op}}$ (same objects, reversed morphisms) satisfies the category axioms.
- (b) Show $(\mathcal{C}^{\text{op}})^{\text{op}} = \mathcal{C}$.
- (c) Show $(\mathcal{C} \times \mathcal{D})^{\text{op}} \cong \mathcal{C}^{\text{op}} \times \mathcal{D}^{\text{op}}$.
- (d) A morphism is an isomorphism in $\mathcal{C}$ iff it is an isomorphism in $\mathcal{C}^{\text{op}}$. Prove this.

Exercise 2.8 (Functors preserve isomorphisms) Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a functor and $f: A \rightarrow B$ an isomorphism in $\mathcal{C}$. Show that $F(f): F(A) \rightarrow F(B)$ is an isomorphism in $\mathcal{D}$, with $F(f)^{-1} = F(f^{-1})$.**Exercise 2.9** (Full and faithful functors)

- (a) Show that the forgetful functor $U: \mathbf{Mon} \rightarrow \mathbf{Set}$ is faithful but not full. (Hint for not-full: exhibit two monoids and a function between their underlying sets that is not a homomorphism.)
- (b) Show that a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is faithful iff it is injective on morphisms between any fixed pair of objects.
- (c) Show that a fully faithful functor reflects isomorphisms: if $F(f)$ is an isomorphism, then so is f .
- (d) $(\star)$ Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be full, faithful and surjective on objects. Show that F is an equivalence of categories (i.e. there exists $G: \mathcal{D} \rightarrow \mathcal{C}$ with $GF \cong id_{\mathcal{C}}$ and $FG \cong id_{\mathcal{D}}$).

Exercise 2.10 (The category **Cat**)

- (a) Show that the identity functor $id_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$ exists for any category $\mathcal{C}$.
- (b) Show that functors compose: if $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$ are functors, so is $G \circ F: \mathcal{C} \rightarrow \mathcal{E}$.
- (c) The diagonal functor $\Delta: \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{C}$ sends $A \mapsto (A, A)$ and $f \mapsto (f, f)$. Verify it is a functor.

Exercise 2.11 $(\star)$ Contravariant powerset is a hom-functor For a set X , let $\mathcal{P}(X) = \text{Hom}_{\mathbf{Set}}(X, 2)$ where $2 = \{0, 1\}$.

- (a) Show there is a bijection between subsets $A \subseteq X$ and characteristic functions $\chi_A: X \rightarrow 2$, $\chi_A(x) = 1$ iff $x \in A$.
- (b) Under this bijection, show that $\text{Hom}_{\mathbf{Set}}(f, 2)(g) = g \circ f$ corresponds to the inverse image $f^{-1}[-]$.
- (c) Conclude that the contravariant powerset functor $\mathcal{P}: \mathbf{Set}^{\text{op}} \rightarrow \mathbf{Set}$ sending $f: X \rightarrow Y$ to $f^{-1}[-]: \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ equals $\text{Hom}_{\mathbf{Set}}(-, 2)$.

3 Natural Transformations and Exponentials

Exercise 3.1 (Exponential objects) Let $\mathcal{C}$ be a category with binary products. Recall that an *exponential* B^A is an object together with $\text{ev}: B^A \times A \rightarrow B$ such that for every $f: C \times A \rightarrow B$ there is a unique $\text{curry}(f): C \rightarrow B^A$ with $\text{ev} \circ (\text{curry}(f) \times \text{id}_A) = f$.

- (a) Verify that in **Set** the exponential B^A is the set of functions $A \rightarrow B$, with $\text{ev}(f, a) = f(a)$.
- (b) Show that if B^A exists then $\text{Hom}(C, B^A) \cong \text{Hom}(C \times A, B)$ naturally in C .
- (c) Deduce that $\text{Hom}(1, B^A) \cong \text{Hom}(A, B)$, where 1 is the terminal object.
- (d) Show that exponentials do not exist in **Mon**. Observe that there is exactly one morphism $1 \rightarrow M$ in **Mon** for any M , so by (c) a necessary condition for B^A to exist is $|\text{Hom}_{\mathbf{Mon}}(A, B)| = 1$.

Exercise 3.2 (Recursion in **Mon**) Let us restate the structural recursion principle from Exercise 1.6 for monoids: For any monoid M , morphism $a: 1 \rightarrow M$, and monoid morphism $h: M \rightarrow M$, there is a unique monoid morphism $f: \mathbb{N} \rightarrow M$ with $f(0) = a$ and $f(n+1) = h(f(n))$.

- (a) Verify this. What is the unique such f explicitly?
- (b) $(\star)$ Show that strong recursion *fails* in **Mon**: exhibit monoids X and M , a monoid morphism $g: X \rightarrow M$, and a monoid morphism $h: M \times X \rightarrow M$ such that no monoid morphism $f: \mathbb{N} \times X \rightarrow M$ satisfies $f(0, x) = g(x)$ and $f(n+1, x) = h(f(n, x), x)$. (Thus, existence of exponentials is indeed a critical step of the proof in Exercise 1.6(c))
(Hint: try $X = M = (2, +_2, 0)$, $g = \text{id}$, $h(m, x) = m +_2 x$, where $+_2$ is addition modulo 2.)